

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Easy

Question

$$(x - 6)^2 + (y - 3)^2 = 81$$

The graph of the given equation in the xy -plane is a circle. What is the length of the radius of this circle?

Answer

- A. 3
- B. 6
- C. 9
- D. 81

Correct Answer: C

Rationale

Choice C is correct. The equation of a circle in the xy -plane can be written as $(x - h)^2 + (y - k)^2 = r^2$, where the center of the circle is (h, k) and the radius of the circle is r . The graph of the given equation, $(x - 6)^2 + (y - 3)^2 = 81$, is a circle in the xy -plane. This equation can be written as $(x - 6)^2 + (y - 3)^2 = 9^2$, where $h = 6$, $k = 3$, and $r = 9$. Therefore, the radius of this circle is 9.

Choice A is incorrect. This is the y -coordinate of the center, not the radius, of the circle defined by the given equation.

Choice B is incorrect. This is the x -coordinate of the center, not the radius, of the circle defined by the given equation.

Choice D is incorrect. This is the value of the radius squared, not the radius, of the circle defined by the given equation.